

## Reproducibility Sheet: Optimization & Learning Report

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Course: CS7641 Machine Learning – Spring 2026

### (a) READ-ONLY Overleaf Link

Overleaf Project Link:

<https://www.overleaf.com/read/xcmwrfphqxtx#897d59>

### (b) GitHub Commit Hash

- **Repository URL:** <https://github.gatech.edu/gt-omscs-ml/cs-7641-2026-spring-urafi3>
- **Commit Hash:** [a4c28022b877f6c3e78d89e24914bef970debc98](#) (or the short version a4c2802)

### (c) Exact Run Instructions

#### Folder Structure

```
project-root/
|
|— data/
|   |— adult.csv
|   |— wine.csv
|
|— outputs/
|   |— figures/
|
|— src/
|   |— __init__.py
|   |— data_loader.py
|   |— models.py
|   |— part1_ro.py
|   |— part2_admin.py
|   |— part3_regularization.py
|   |— part4_optimizers.py
|   |— paths.py
```

```
|  └─ ro_optimizers.py
|  └─ utils.py
|
└─ requirements.txt
└─ README.md
```

## Setup Instructions

1. Place the datasets in the data/ directory.

Required datasets:

- Adult Income Dataset
- Wine Quality Dataset

2. Install Python dependencies:

```
pip install -r requirements.txt
```

3. Run the files for each of the requirements:

```
python src/part1_ro.py
```

```
python src/part2_admin.py
```

```
python src/part3_regularization.py
```

```
python src/part4_optimizers.py
```

This script executes all experiments required for the Optimization & Learning report:

- Randomized Optimization experiments
- Optimizer ablation experiments
- Regularization studies
- Comparison studies

Outputs (plots and results) will be saved in the outputs/ directory.

## (d) EDA Summaries

### Adult Income Dataset

The dataset contains 48,842 instances after cleaning, with 14 features split between 6 numerical and 8 categorical variables. The task is binary classification: predicting whether income is  $\leq \$50K$  or  $> \$50K$ . There's a moderate class imbalance with 75% earning  $\leq \$50K$  (36,631) and 25% earning  $> \$50K$  (12,211), giving roughly a 3:1 ratio.

For numerical features, age ranges from 17-90 (mean ~38.6) and hours-per-week ranges 1-99 (mean ~40.4). Education-num captures years of education (1-16). Both capital-gain (max 99,999) and capital-loss (max 4,356) are heavily right-skewed with most values at zero, indicating few people have significant investment income. The fnlwgt (final weight) variable has wide-ranging values.

Categorical features include workclass (8 levels: Private, Self-employed, Government, etc.), education (16 levels from Preschool to Doctorate), marital-status (7 categories), occupation (14 types including Techsupport,

Sales, Exec-managerial), relationship (6 types), race (5 categories), sex (Male/Female), and native country (United-States plus 40+ others).

Missing values appear in workclass, occupation, and native-country. I handled these by creating a 'missing' category for categoricals and using median imputation for numericals.

### **Wine Quality Dataset**

This dataset has 6,497 wine samples with 12 numerical features plus a binary wine type indicator (red=0, white=1). The target is quality rating from 3-9, though the distribution is highly imbalanced: most wines cluster around quality 5 (33%) and 6 (44%), with quality 7 at 17%. The extremes are rare—only 30 samples rated 3, and just 5 rated 9.

The 11 physicochemical features include acidity measures (fixed: 3.8-15.9, volatile: 0.08-1.58, citric: 0-1.66), residual sugar (0.6-65.8, right-skewed), chlorides (0.009-0.611), sulfur dioxide levels (free: 1-289, total: 6-440), density (0.987-1.039), pH (2.72-4.01), sulphates (0.22-2.0), and alcohol content (8.0-14.9%). Most features have reasonable ranges except residual sugar, which shows high variability.

There are no missing values in this dataset. This dataset is used for:

- Randomized Optimization experiments
- Optimizer ablation experiments
- Regularization studies
- Comparison studies